Decorator Pattern

The Decorator Pattern is a **structural** design pattern that lets you **add new behaviors** to **objects dynamically** by wrapping them in decorator objects. Instead of modifying the original class or creating subclasses for every combination, you wrap objects in layers that add functionality

Problem Statement

You're building a **Bar Ordering System**. Customers order **drinks** with add add-ons:

Base Spirits:
- Whiskey ($5.00)
- Vodka ($5.00)
- Rum ($4.50)

Add-ons (mixers):
1. Ice - $0.00 (can add multiple for "extra ice")
2. Coke - $1.00
3. Soda - $1.00
4. Lime - $0.50
5. Orange Juice - $1.50

Customers decide add-ons **at the bar**, after the drink is poured. They can add more anytime ("actually, add some lime too").

How do you add behaviors to existing objects dynamically at runtime?

$ 5

Whiskey

5+0 = $5

Whiskey
+
Ice

5+0+1 = $6

Whiskey
+
Ice
+
Coke

5+0+1+1 = $7

Whiskey
+
Ice
+
Coke
+
Soda

Approach 1 (Subclass for every Combination)

```
class Whiskey { }
class WhiskeyWithIce extends Whiskey { }
class WhiskeyWithCoke extends Whiskey { }
class WhiskeyWithIceAndCoke extends Whiskey { }
```

Class explosion -
- With N add-ons, you need 2^N classes for all combinations

Decisions at compile time -
- No Run Time Modifications

Approach 1 (Subclass for every Combination)

```
class Whiskey { }
class WhiskeyWithIce extends Whiskey { }
class WhiskeyWithCoke extends Whiskey { }
class WhiskeyWithIceAndCoke extends Whiskey { }
class WhiskeyWithIceAndLime extends Whiskey { }
class WhiskeyWithIceAndCokeAndLime extends Whiskey { }
class WhiskeyWithIceAndCokeAndLimeAndSoda extends Whiskey { }
```

Class explosion -
- With N add-ons, you need 2^N classes for all combinations

Decisions at compile time -
- No Run Time Modifications

No "extra ice" -
- Would need `WhiskeyWithDoubleIce` , `WhiskeyWithTripleIce`

Approach 2 (Boolean Flags in the Whiskey Class)

```
class Whiskey{
    private boolean hasIce;
    private boolean hasCoke;
    private boolean hasLime;
    private boolean hasSoda;

    public void addIce() { this.hasIce = true; }
    public void addCoke() { this.hasCoke = true; }

    public double getCost() {
        double cost = 5.00;
        if (hasIce) cost += 0.00;
        if (hasCoke) cost += 1.00;
        if (hasLime) cost += 0.50;
        if (hasSoda) cost += 1.00;
        return cost;
    }
}
```

- **Violates OCP** - New mixer = modify class [Mixers are tighlty coupled]
- **No multiples** - Boolean can't represent "extra ice"
- **Doesn't extend** - What about Vodka?

```
Whiskey whiskey = new Whiskey();
whiskey.addIce();
whiskey.addCoke();
```

whiskey.getCost()

Decorator Pattern for this problem

add new behaviors to existing objects by wrapping them in

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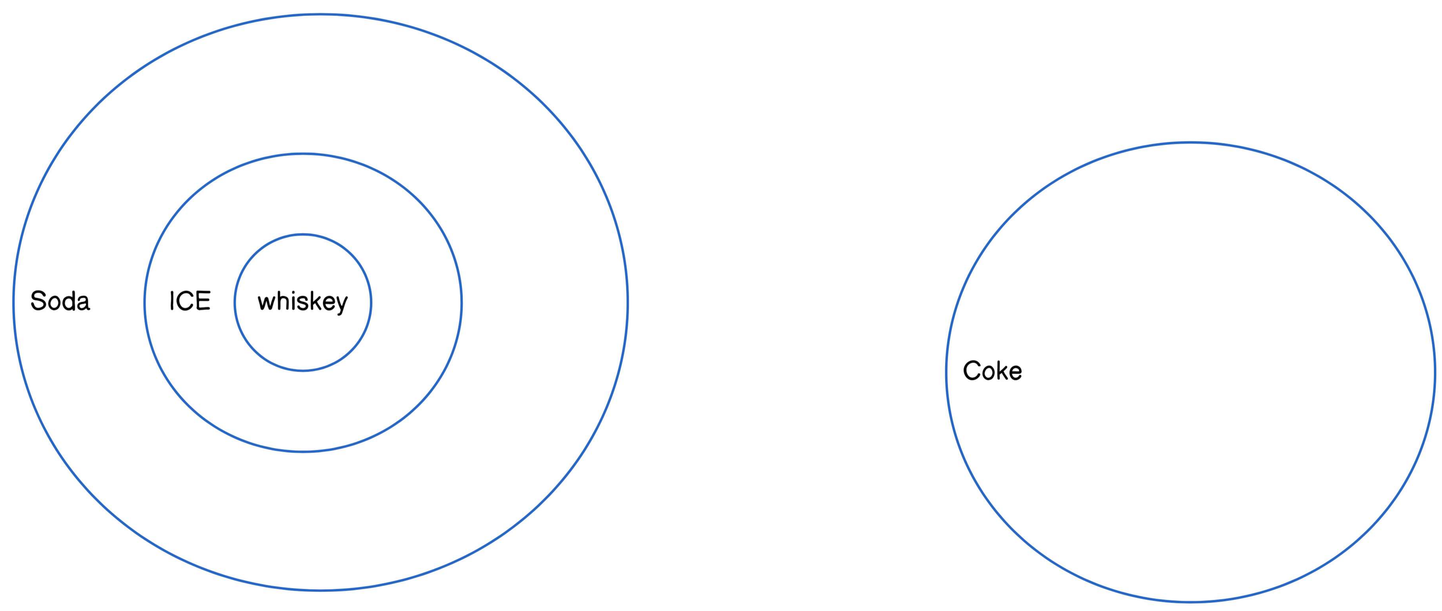
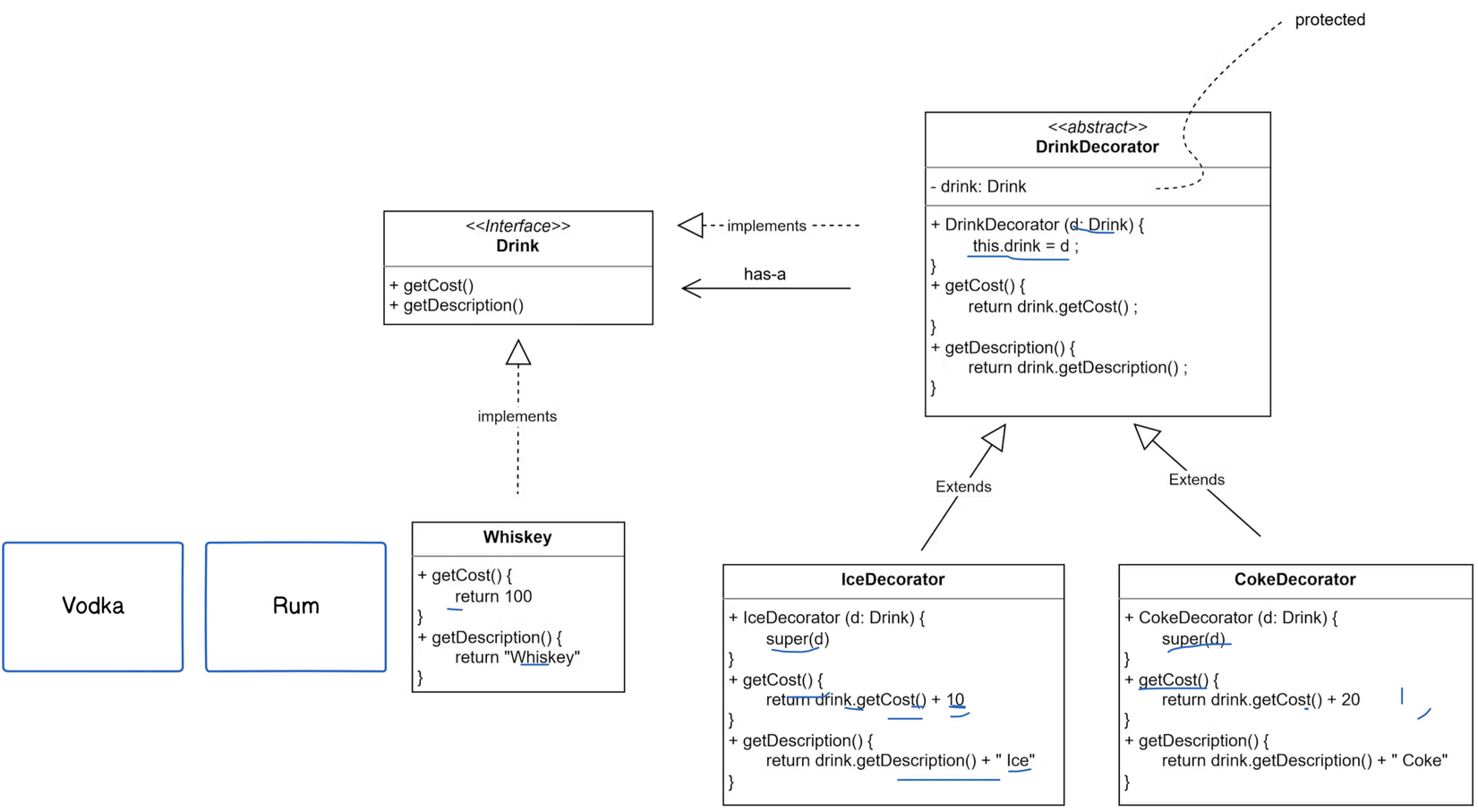
add new behaviors to existing objects by wrapping them in decorator objects




```
Drink order = new Whiskey(); //base drink
order = new IceDecorator(order);
order = new CokeDecorator(order);
order = new LimeDecorator(order);
```

Decorator Pattern Drink Example Pseudo Code

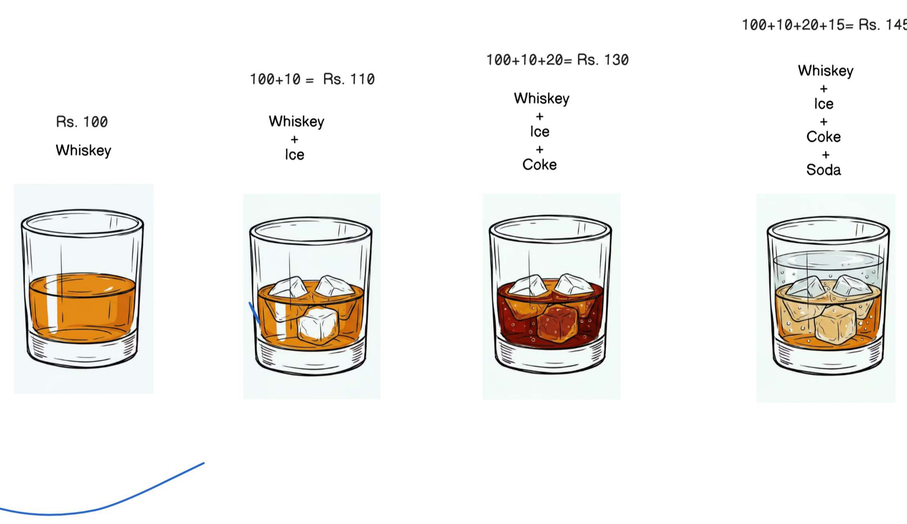



```
Drink order = new Whiskey();

order = new IceDecorator(order);

order = new StrawDecorator(order);

order.getCost();
order.setColor();


order = new CokeDecorator(order);
order = new SodaDecorator(order);
```

```
class StrawDecorator extends DrinkDecorator{

  private String color="red";

  public strawDecorator(Drink d){
    super(d);
  }

  public String getStrawColor(){
    return color;
  }

  public String setColor(String col){
    this.color = col;
  }

}
```

Problems so far:

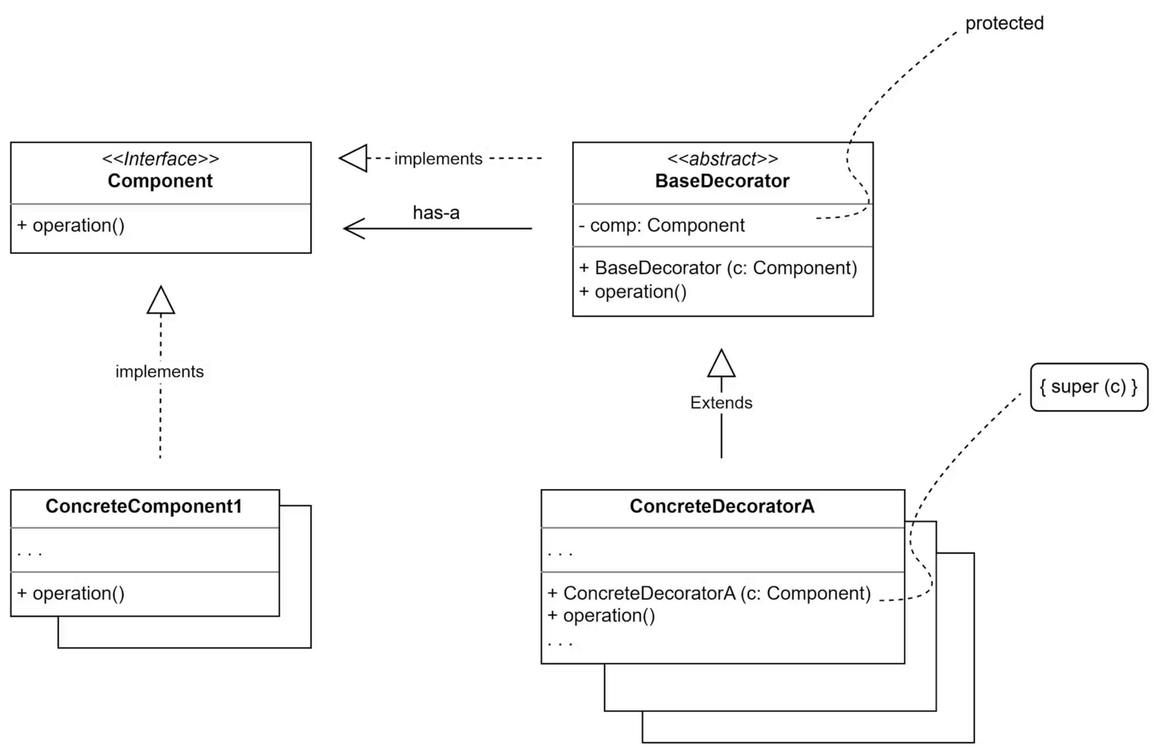
1) Compile Time decisions
2) No Multiples
3) Extension
4) Class Explosion
5) OCP (new mixer)



Not necessary that concrete decorator implements all the get cost and get description
E.g. StrawDecorator

```
class StrawDecorator extends DrinkDecorator{

  private String color="red";

  public strawDecorator(Drink d){
    super(d);
  }

  public String getStrawColor(){
    return color;
  }

  public String setColor(String col){
    this.color = col;
  }

}
```

Decorator Pattern Standard UML Diagram




```
Component c = new ConcreteComponent1();
c = new ConcreteDecoratorA(c);
c = new ConcreteDecoratorB(c);
c = new ConcreteDecoratorC(c);
```